O-I-533C
August 28, 1970
SUPERSEDING
Int. Fed. Spec. O-I-00533B (GSA-FSS)
May 7, 1969 and
Fed. Spec. O-I-533A
February 13, 1958

FEDERAL SPECIFICATION

INSECTICIDE, LINDANE, LIQUID EMULSIFIABLE CONCENTRATE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The liquid insecticide concentrate covered by this specification is designed to be diluted with the specified amount of water and used as a residual spray.

1.2 Classification.

1.2.1 Type and grade. The liquid insecticide concentrate covered by this specification shall be of one type and grade.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-X-916 - Xylene (for Use in Organic Coatings).
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.
PPP-D-729 - Drums: Metal, 55-Gallon (for Shipment of Noncorrosive Material).
PPP-P-704 - Pails, Metal: (Shiping, Steel 1 through 12 Gallon).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 141/4082.1 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6840

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Laws and Regulations:

Federal Insecticide, Fungicide and Rodenticide Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D-92-66 - Flash and Fire Points by Cleveland Open Cup.
D-86-67 - Distillation of Petroleum Products.
D-1500-64 - ASTM Color of Petroleum Products.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Nongovernmental:

Official Methods of Analysis of the Association of Official Analytical Chemists; Tenth Edition (1965).

(Application for copies should be addressed to the Association of the Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent, 1616 P Street, N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Composition. Finished insecticide, emulsifiable concentrate shall be composed of emulsifying agent, and solvent and shall contain 12 grams (gm.) of lindane per 100 milliliters (ml.) approximately one pound of lindane per gallon of solution at 20° C. (68° F.) as determined in 4.5.5.

3.1.1 Lindane. The lindane shall be the gamma isomer of 1, 2, 3, 4, 5, 6-hexachlorocyclohexane of a purity not less than 99 percent. It shall be a white crystalline substance having a melting point of 112° to 113° C., when determined as specified in 4.5.1.

3.1.2 Emulsifying agent. The emulsifying agent shall be completely soluble in the finished insecticide concentrate; shall be stable in the presence of lindane; shall not cause breakdown of the lindane; and when in the finished concentrate shall meet the emulsification test described in 4.5.2.

3.1.3 Solvent. The solvent shall consist principally of aromatic petroleum derivatives having the properties shown in table I when the solvent is, tested as specified in table II. The solvent may also contain up to 12.5 percent, maximum of xylene meeting the requirements of grade A-ten-degree xylene according to TT-X-916 provided the flash point of the finished formulation is not lowered below 140° F.

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TABLE I. Solvent properties

Flash point (Cleveland open cup)	
Minimum °F.	140
Distillation range:	
Initial boiling point, minimum °F.	300
End point, maximum °F.	590
Color - ASTM - Union colorimeter, maximum	2.0

3.2 Appearance. The finished product shall be a clear, homogeneous solution, free of sediment, and there shall be no separation of solid or liquid phase when tested for homogeneity as specified in 4.5.4.

3.3 Emulsion stability. The emulsion formed with both hard and soft water, shall show not more than 2 ml. of creamy layer separation and no oil separation when tested after formation and reformation as specified in 4.5.2.

3.4 Flash point. The flash point of the finished insecticide concentrate shall be not less than 140° F. when tested as specified in 4.5.3.

3.5 Color and staining. The insecticide concentrate shall give a color reading of not more than 2.0 on the Union colorimeter when tested as specified in 4.5.3. There shall be no residual stain when the emulsifiable concentrate is tested as specified in 4.5.6.

3.6 pH. The finished product shall have a pH between 5.0 and 7.5 when tested as specified in 4.5.7.

3.7 Residual odor. There shall be no more than a slight mild residual odor when the emulsifiable concentrate is tested as specified in 4.5.6.

3.8 Water content. The finished product shall not contain more than 0.15 percent water when tested as specified in 4.5.8.

3.9 Workmanship. The product shall be clean and uniform and free of any defects which might impair its utility.

3.10 Registration. The insecticide formulation shall be registered with the U. S. Department of Agriculture for compliance with the Federal Insecticide, Fungicide and Rodenticide Act.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.2 Examination of the end item. Examination of the end item shall be made in accordance with classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size, shall be expressed in units of filled unit containers of the same capacity for examinations in 4.2.2.1 and 4.2.2.2 and in units of shipping containers for the examination in 4.2.2.3. A lot shall consist of insecticide produced using the same raw material lots under the same operating conditions.

4.2.2.1 Examination for visual defects. The sample unit for this examination shall be one gallon of material. Only one sample unit shall be taken from any one primary container. Each sample unit selected shall be examined for defects in appearance with respect to form, uniformity and presence of foreign particulate matter.

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4.2.2.2 Examination of filled containers for net contents. The sample unit for this examination shall be one filled unit container. The average net contents per container for all sample units examined shall be not less than quantity specified.

4.2.2.3 Examination for preparation for delivery. An examination shall be made to determine that packaging, packing, and marking comply with the requirements of section 5 of this specification. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging	Container not as specified. Container damaged, leaking, or otherwise defective. Closure not as specified.
Packing (as applicable)	Not level specified, not in accordance with contract requirements. Arrangement of units per shipping container not as specified. Container not as specified, closure not accomplished by specified methods or materials. Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability. Inadequate application of components, such as incomplete closures of containers, container flaps, loose or inadequate strapping, bulged or distorted containers.
Count (as applicable)	Less than specified or indicated quantity of primary containers per shipping container.
Marking	Omitted, incorrect, illegible, incomplete or not in accordance with contract requirements. Precautionary markings missing or not as specified.

4.2.3 Inspection levels and acceptable quality levels (AQL's) for examination. The inspection levels for determining the sample size, and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1	I	2.5
4.2.2.2	S-2	N/A
4.2.2.3	S-2	2.5

4.3 Testing of the end-item. The end-item shall be tested in accordance with the tests specified in 4.5 for conformance to the specified requirements. The sample unit for tests shall be one quart of material. The sample size shall be S-1 with no less than three sample units randomly selected from the lot. There shall be no failure to meet the specified requirements.

4.4 Tests. The sample specimens selected in accordance with 4.2.3 shall be subjected separately to the tests specified in 4.5. If either specimen fails one or more of these tests, the lot shall be rejected.

4.5 Test methods.

4.5.1 Melting point. The melting point shall be determined by the capillary tube method, using a Thiele-Dennis, or equivalent, type melting tube, heated at a rate not to exceed 0.5° C. per minute. Initial reading shall be taken when first appearance of liquid is observed and final reading, when the sample is completely liquefied.

4.5.2 Emulsion stability (at 80° ± 5° F.).

4.5.2.1 Apparatus. The test emulsion shall be prepared in a 250 ml. Griffin low-form beaker (approximately 85 mm. high and 65 mm. diameter). A T-shaped stirrer shall be used which is made of metal or glass and consist of a rod 10 mm. in diameter and 50 mm. long, attached at right angles to the end of a shaft of 5 mm. in diameter. The stirrer shall be raised off of the bottom of the beaker just enough to permit free rotation.

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4.5.2.2 Emulsification. Place 95 ml. of hard water (4.5.2.3) at 80° ± 5° F. in the beaker and stir at 1000 ± 50 r.p.m. Add 5.0 ml. of the emulsion concentrate from a pipette and continue stirring for one minute. Pour the emulsion thus formed into a 100 ml. graduated glass-stoppered cylinder, note the time, and set aside for exactly 30 minutes. Immediately after that period examine the emulsion carefully under strong transmitted light (100 watt) for signs of separation of oil or top or bottom creaming and record the percentage separation, if any, by volume. Allow the emulsion to stand at the test temperature for 24 hours. Then reform the emulsion by inverting the stoppered cylinder through 30 complete cycles. Exactly 30 minutes after the emulsion has been reformed, examine again under strong light and record the results. The emulsification tests shall be repeated using soft water (4.5.2.4) in place of hard water.

4.5.2.3 Hard water. The hard water specified in 4.5.2.2 shall have the following composition, an is designed to provide a hardness of 342 p.p.m. calculated as calcium carbonate.

$CaCl_2$ (anhydrous)------- 0.3037 gram
$MgCl_2.6H_2O$ ------------ .1388 gram
H_2O (distilled) to make- 1 liter

4.5.2.4 Soft water. The soft water specified in 4.5.2.2 shall be prepared by diluting one part of hard water (4.5.2.3) with five parts of distilled water. This forms a test water of 57 p.p.m. hardness calculated as calcium carbonate.

4.5.3 Solvents. Tests on solvents shall be determined in accordance with ASTM designations as specified in table II. The test method of flash point and color are also applicable to the finished insecticide concentrate.

TABLE II. Solvent Tests

Test	Method
Flash point (Cleveland open cup)	D-92
Initial boiling point	D-86
End point	D-86
Color - ASTM - Union colorimeter	D-1500

4.5.4 Homogeneity. Examine the sample of liquid concentrate visually by transmitted and reflected light for clarity, homogeneity, and undissolved or foreign matter. A slight haze or turbidity is acceptable. The test shall be repeated after storage for 24 hours at 20° F., seeding with a crystal or two of lindane and restoring at 20° F. for 96 hours.

4.5.5 Lindane content.

4.5.5.1 Partition chromatography method. The apparatus, reagents, preparation of column and determination to be used are those described in sections 4.209, 4.210, 4.212, and 4.213 of the Official Methods of Analysis of the Association of Official Analytical Chemists, Tenth Edition (1965).

4.5.5.2 Preparation of sample. Pipette at 68° F., 20 ml. of the concentrate into a 100 ml. volumetric flask, add 1 ml. of the dye solution and dilute to the mark with the mobile solvent.

4.5.6 Staining properties and residual odor. A 6-inch square of bleached cotton sheeting shall be immersed in the diluted concentrate (4.5.2.2) until thoroughly wetted, then wrung to remove excess liquid, hung up to dry for 48 hours in a well-ventilated room. At the end of 48 hours, the treated section of the sheeting shall be examined and compared to a similar untreated piece of sheeting for staining and residual odor.

4.5.7 pH. The pH value at 25° C. shall be determined by any standard means on aqueous test emulsion as prepared in 4.5.2.2.

4.5.8 Water content. This method involves the direct determination of water by titration with a solution of iodine, sulfur dioxide and pyridine in methanol (Karl Fischer Reagent). The procedure shall be as specified in Fed. Test Method 141/4082.1 with substitution of emulsifying agent, or finished product, as applicable, for paint or varnish.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Insecticide shall be furnished in 1-quart, 5-gallon or 55-gallon quantities, as specified (see 6.2).

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5.1.1.1 One quart quantities of insecticide shall be furnished in bottles conforming to group A, class 1, type a, style 1, grade 1, closure A, seal A or B in accordance with PPP-C-186.

5.1.1.2 Five-gallon quantities of insecticide shall be furnished in metal pails conforming to type I, class 8 of PPP-P-704.

5.1.1.3 Fifty-five gallon quantities of insecticide shall be furnished in metal drums conforming to type II of PPP-D-729.

5.1.2 Level B. Insecticide shall be packaged as specified in 5.1.1 above.

5.1.3 Level C. Insecticide shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Insecticide packaged as specified in 5.1.1.1, shall be packed in accordance with PPP-C-186.

5.2.1.2 Insecticide packaged as specified in 5.1.1.2 and 5.1.1.3 will require no additional packing.

5.2.2 Level B.

5.2.2.1 Insecticide packaged as specified in 5.1.1.1, shall be packed in accordance with PPP-C-186.

5.2.3 Level C. Insecticide in quantities as specified (see 6.2) packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

6. NOTES

6.1 Approved use. The insecticide formulation described by this specification is approved for use listed in the label as registered with the U. S. Department of Agriculture.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Size container required (see 5.1).
(d) Quantity required (see 5.2.3)

MILITARY INTEREST:

Custodians:

Army - MU
Air Force - 68

User activity:

Navy - YD

Preparing activity:

GSA-FSS

CIVIL AGENCIES INTEREST:

GSA-FSS
VA
AGR

Military coordinating activity:

Army - (GL)

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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